AP CALCULUS AB

Unit 1 • Unit 1 Fix-Up • Limits, Continuity, and Differentiability • Answer key

Unit 1 Fix-Up • Limits, Continuity, and Differentiability

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Part A • Multiple choice

Question 1

NO CALCULATOR

$\lim_{x \to -4} \dfrac{x^2 - 16}{x + 4}$ is

(A) 0

(B) -8

(C) 8

(D) nonexistent

Solution

Answer: (B).

Question 2

NO CALCULATOR

$\lim_{x \to \infty} \dfrac{2x - 7}{x^2 + 1}$ is

(A) 0

(B) 2

(C) -7

(D) ∞

Solution

Answer: (A).

Question 3

NO CALCULATOR

$\lim_{x \to 3^+} \dfrac{-1}{x-3}$ is

(A) ∞
(B) -1
(C) $-\infty$
(D) 0

Solution
Answer: (C).

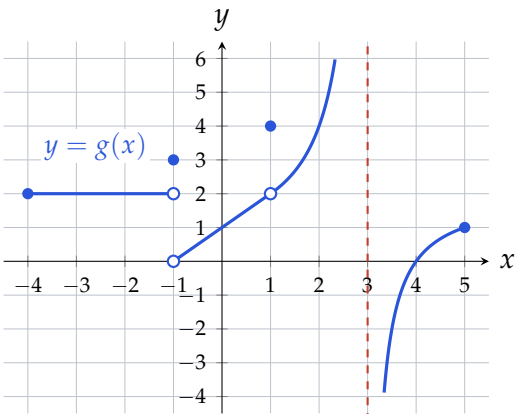
Question 4

NO CALCULATOR

$\lim_{x \to 0} \dfrac{\sin x}{4x}$ is

(A) 4
(B) $\frac{1}{4}$
(C) 1
(D) 0

Solution
Answer: (B).



Question 5

NO CALCULATOR

Using the graph of g, which limit does not exist?

(A) $\lim_{x \to 1} g(x)$
(B) $\lim_{x \to 4} g(x)$
(C) $\lim_{x \to -1} g(x)$
(D) $\lim_{x \to -2} g(x)$

Solution
Answer: (C).

Question 6

NO CALCULATOR

Using the graph of g, what type of discontinuity is at $x = 3$?

(A) removable
(B) jump
(C) infinite
(D) none

Solution
Answer: (C).

Question 7

NO CALCULATOR

For which k is $f(x) = \begin{cases} 2x + k, & x < 1 \\ kx^2 + 3, & x \geq 1 \end{cases}$ continuous?

(A) 1
(B) -1
(C) It is continuous for every k.
(D) No value of k works.

Solution
Answer: (D). $2 + k = k + 3$ is impossible.

Question 8

NO CALCULATOR

Which function must have an absolute maximum on $[0, 2]$?

(A) $\frac{1}{x-1}$
(B) $\frac{1}{x+1}$
(C) $\tan(x)$ with x in radians
(D) $\frac{1}{x}$

Solution
Answer: (B). Continuous on $[0, 2]$. (tan is undefined at $\frac{\pi}{2} \approx 1.57$.)

Question 9

NO CALCULATOR

$\lim_{x \to 5} \dfrac{x^2 - 25}{x - 5}$ represents $f'(5)$ for which f?

(A) $f(x) = x^2$
(B) $f(x) = x - 5$
(C) $f(x) = x^2 - 25$ at $x = 0$
(D) $f(x) = 2x$

Solution
Answer: (A). $\frac{f(x)-f(5)}{x-5}$ with $f(5) = 25$; value 10.

Question 10

NO CALCULATOR

For $f(x) = x^2$ on $[2, 6]$, the MVT gives $c =$

(A) 3
(B) 5
(C) 4
(D) 8

Solution
Answer: (C). Secant slope $\frac{36-4}{4} = 8 = 2c$.

Part B • Free response

Question 11 (Free response)

NO CALCULATOR

Let $h(x) = \dfrac{x^2 - x - 2}{x^2 - 1}$.

(a) Find and classify each discontinuity, with limits.
(b) Find $\lim_{x \to \pm\infty} h(x)$.
(c) What value of h at the removable discontinuity makes h continuous there?

Solution
$h = \frac{(x-2)(x+1)}{(x-1)(x+1)}$. (a) $x = -1$: removable ($\lim = \frac{-3}{-2} = \frac{3}{2}$). $x = 1$: infinite ($\lim_{x \to 1^-} = +\infty$, $\lim_{x \to 1^+} = -\infty$). (b) 1. (c) $\frac{3}{2}$.

Question 12 (Free response)

CALCULATOR

Let $f(x) = x^3 - 4x + 1$.

(a) Use the IVT to show f has a zero in $(0, 1)$.
(b) Find the value(s) of c in $(0, 2)$ guaranteed by the MVT, using $f'(x) = 3x^2 - 4$.
(c) Use a calculator to find all zeros of f in $(0, 2)$.

Solution
(a) Continuous; $f(0) = 1 > 0 > -2 = f(1)$.
(b) $\frac{f(2)-f(0)}{2} = \frac{1-1}{2} = 0 = 3c^2 - 4$: $c = \frac{2}{\sqrt{3}} \approx 1.155$.
(c) $x \approx 0.254$ and $x \approx 1.861$.